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LIVE YOGA POSE DETECTION BY USING IMAGE PROCESSING AND DEEP LEARNING

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Abstract: This article presents a method for reliably assessing different yoga poses using deep learning algorithms. We also suggest a better scoring mechanism for this system, one that can be used for all yoga poses. Our app is assessed on its robustness as well as various yoga positions performed in various settings. To aid in the self-learning of yoga, we suggest a pose identification approach for yoga position assessment in this system. The method uses the PC camera's multi-part detection to first identify a yoga stance. For the purpose of recognizing yoga poses in real-time films, a hybrid deep learning model is created using LR. LR is utilized to extract features from each frame's critical points.

Keywords: *Deep Learning, Linear Regression, Machine Learning*

1. Introduction

The field of computer vision has difficulty in estimating human position. It deals with the localization of human joints in an image or video to form a skeletal representation. To automatically detect a person's pose in an image is a difficult task as it depends on a number of aspects such as scale and resolution of the image, illumination variation, background clutter, clothing variations, surroundings, and interaction of humans with the surroundings. Exercise and fitness are an application of posture estimation that has drawn the attention of many researchers in this subject. One form of exercise with intricate postures is Yoga which is an age-old exercise that started in India but is now famous worldwide because of its many spiritual, physical, and mental benefits. The problem with Yoga however is that, just like any other exercise, it is of utmost importance to practice it correctly as any incorrect posture during a Yoga session can be unproductive and possibly detrimental. Due to this, an instructor is required to oversee the session and adjust each participant's posture. Since not all users have access or resources to an instructor, an artificial intelligence-based applications might be used to

identify Yoga Poses and provide personalized feedback to help individuals improve their form. In recent years, human pose estimation has benefited greatly from deep learning, and huge gains in performance have been achieved. Deep learning approaches provide a more straightforward way of mapping the structure instead of having to deal with the dependencies between structures manually using deep learning to identify 5 exercise poses: pull-up, Swiss-ball hamstring curl, push-up, cycling and walking. However, using this method for Yoga poses is a relatively newer application. Deep learning has substantially helped human pose estimation in recent years, and significant performance improvements have been made. What is deep learning? How can deep learning be applied to Yoga pose classification in real-time? This project uses references from conference proceedings, published papers, technical reports, and journals. The first section of the project talks about the history and importance of Yoga. The second section discusses posture estimation and goes into greater length to explain the various types of pose estimation methods, as well as learning-based (deep learning) and exemplar-based discriminative methods.

2. Literature Review

Real-time human monitoring has received a lot of attention for a range of applications. The suggested study seeks to analyze the existing literature and contribute to improving the findings that are now attainable in the area of computer vision for estimating human stance. A Deep Learning-based system has been developed in an effort to effectively identify yoga poses and serve as a user's substitute for a trainer by providing them with appropriate feedback. In a comparison of

machine learning and deep learning methods for estimating yoga poses, Kothari et al. [2] conducted an experiment involving both technologies and evaluated the effectiveness of Neural Networks. The study found that the hybrid CNN-LSTM model performed best and produced the fewest misclassifications. Kadbhane et al.'s [11] Microsoft Kinect was used to record video data. A skeletal tracking tool for Kinect can recognize up to twenty body joints. They calculated using the ten joint sites they had picked. By accumulating information about the joint sites from human posture, they developed the reference Page 5/25 framework for each yoga position. By calculating the angles between all the vectors connecting any two joint sites, they derived the cosine similarity of vectors. When the estimated divergence exceeds a predetermined threshold, the pose is deemed wrong. The overall accuracy attained was not mentioned in the paper. ExNet is a multilayer convolutional neural network (CNN) that was proposed by S. Haque et al. [12]. They made use of a dataset of 2000 photographs of people in exercise stances, which was separated into five classes: push-ups, pull-ups, cycling, swiss ball hamstring curls, and walking. The Adam optimizer and automated learning rate reduction approach were utilized in the model. ExNET was able to classify a 2D human exercise stance from the dataset with 82.68% accuracy after 50 iterations. The model has overfitting issues and requires greater hyperparameter tuning. The dataset used by Agarwal et al. [13] included 5500 photos of 10 different yoga poses. They suggested a system in which the user's skeleton is initially identified using the tf-pose estimation algorithm. The last step involved using and contrasting six different machine learning models: Decision Tree, Random Forest, Logistic Regression, Naive Bayes, SVM, and KNN. With a 99.04% accuracy rate, the Random Forest classifier had the best performance. Anilkumar et al. [14] proposed a method in which the mediapipe library performs the geometric analysis based on the frames collected from the camera whenever a stance is made by the user in front of his camera. The results of the geometric analysis are based on the angles between various joints. The

the framework using support vector machines, convolutional neural networks, and Convolutional

accurate angles saved in the database for the particular yoga posture are then compared to the calculated angles. The user receives feedback in the form of text or audio if the difference between these two angles exceeds the predetermined threshold [14]. A system that can recognize poses associated with four different types of typical human activity and recognize human action in both the second and third dimensions was proposed by Luvizon et al. [15]. The study used the multitask convolutional neural network to analyze data from four different sources, including Human 3.6M, NTU, Penn Action, and MPII. Chen et al. [16], who employed a dataset with more than 100 films of exercises being performed correctly and badly, suggested another model to improve posture when exercising. The model was trained using OpenPose, and the scientists defined 18 key points to precisely identify posture changes. Four separate exercises, the front raise, standing shoulder press, bicep curl, and shoulder shrug were used to complete the assignment. Using deep learning techniques, Zou et al. [17] created a smart fitness trainer based on the ideas of pose detection and estimate. The work has made use of a regional multi-person pose estimation framework. The key points, which are the human joints within each frame, were located using AlphaPose. After linking them, the figure is compared to the optimum poses to provide tips to the user on how to perform the exercises properly. Tadasana, Shavasana, Bhujangasana, Trikonasana, Padmasana, and Trikonasana were included in the dataset used by Yadav et al. [18] to accurately develop a real-time deep learning network using OpenPose to extract the key points from the provided movies. With input obtained framewise, the system could achieve an accuracy of 99.04%, whereas detection based on a survey of the pre-selected 45 edges produced a performance measure of 99.38%. The work was later evaluated on twelve people and found to be extremely accurate with a

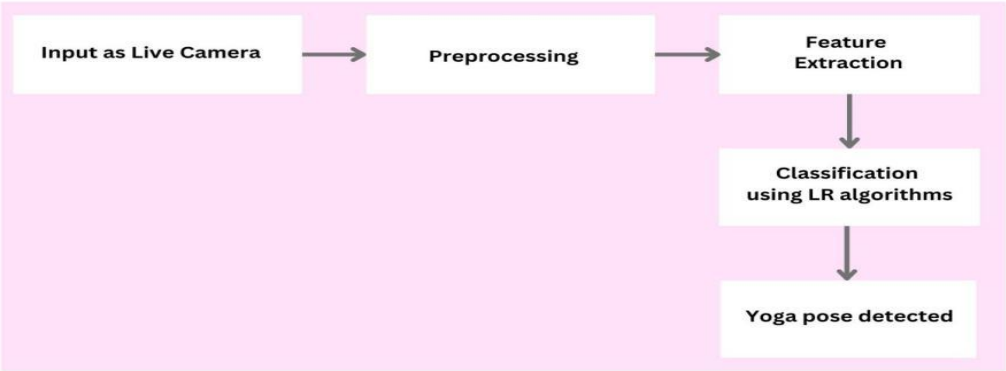

Figure 1. Proposed System Block Diagram

98.92% success rate. The deep learning method was also used by Thoutam et al. [19] to train the model to recognize yoga positions. Multilayer Perceptron was used to classify the features into one of the six poses: corpse pose, tree pose, mountain pose, triangle pose, lotus pose, and cobra pose based on the angles between the twelve key points. This classification had an accuracy of 99.58%.

3. Methodology

Artificial intelligence has a subfield called machine learning, which describes the capacity of IT systems to solve issues on their own by identifying patterns in databases.In other words, using existing algorithms and datasets, machine learning helps IT systems to identify patterns and create appropriate solution concepts. As a result, artificial knowledge is created by machine learning using experience. The software needed human intervention in the past to be able to produce solutions on its own. For instance, the systems must be

prepared in advance with the necessary algorithms, data, and analytic rules for identifying patterns in the data stock. Following the completion of these two processes, the system can carry out the following functions using machine learning:

- data Calculating probabilities for specific
- autonomously optimizing processes based on recognized patterns
- Finding, extracting and summarizing relevant
- data Making predictions based on the analysis
- results Adapting to certain developments

1. Live Camera Input: At this step in the pipeline, a camera's live video stream is taken in as input. It might be a webcam, a camera on a smartphone, or any other gadget that can record video.
2. Preprocessing: In order to enhance the data's quality and usability, the live camera input frequently has to be processed. To improve the clarity and stability of the video feed, preprocessing techniques might be used to resize the frames, alter brightness or contrast, remove noise, or use image stabilization methods.
3. Feature Extraction: After preprocessing, each video frame must then have pertinent features extracted from it. Features for yoga pose detection could be generated from critical places on the human body, including the positioning of joints or body parts. For feature extraction, a variety of computer vision techniques can be applied, such as optical flow analysis, posture estimation algorithms, and keypoint identification.
4. Classification using LR Algorithm: After features are extracted, the pipeline uses a classification algorithm to ascertain the yoga stance based on the features that were extracted. You mentioned applying a logistic regression (LR) approach in this situation. For binary classification tasks, a popular approach is logistic regression. It forecasts the class probabilities for each frame by modeling the relationship between the characteristics and the target variable, in this example, various yoga positions.
5. Yoga Pose Detection: The pipeline can recognize and detect the yoga stance by examining the expected probability after the classification algorithm has made its prediction for each frame. The pose with the highest probability might be taken into account as the detected pose, or extra logic could be used to combine temporal data from numerous frames to increase accuracy and stability.

This block diagram shows the fundamental steps of a pipeline used to detect yoga poses in real-time from a live camera feed. It processes the video frames, extracts the pertinent information, classifies the postures using logistic regression, and then detects the yoga poses based on the classification outcomes. It integrates computer vision techniques with machine learning algorithms.

Our main objectives are:

1. To identify human action using image processing.

2. To detect identification of the person's Yoga, pose and correct them.

3. To study and extract the optical flow of image data and propose a Linear Regression model to deal with the problem.

4. To learn a deep learning model correctly tries the different poses so as to accurately classify them.

5. To extract Open Pose (a pre-trained model) key points of the human joint locations from the images and then train the deep learning model on these key points.

6. Our goal is to train a custom deep-learning model to detect whether a person's Yoga poses are correct or not

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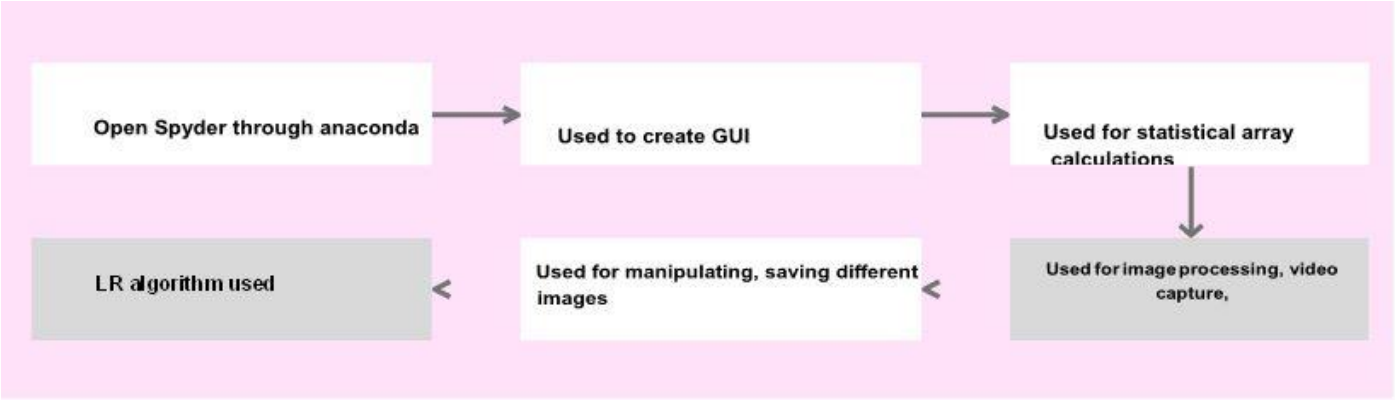

Figure 2. Use case diagra

Open Spyder through Anaconda: Spyder is a Python-integrated development environment (IDE) that can be opened using Anaconda. It is used to write and modify Python code, and it offers a variety of tools for testing and debugging programs. Spyder and a number of additional Python libraries are included with the Python distribution known as Anaconda.

Create a GUI: A GUI (graphical user interface) is a user interface that communicates with users visually through the use of windows, buttons, and menus. Python programs can use Spyder to build GUIs.

Use Spyder to compute statistical arrays: Spyder can be used to compute statistical arrays. A group of data points can be stored in an array, which is a type of data structure.

Use Spyder for image processing and video capture: Spyder is capable of both tasks. Digital image alteration is known as image processing. The act of recording video from a camera is known as video capture.

Use Spyder to modify and save various photos: Spyder may be used to modify and save various photographs. Image manipulation is the technique of altering an image's appearance. The act of saving an image to a file is known as image saving.

Used the LR algorithm: A machine learning approach called LR (logistic regression) can be applied to categorization tasks. Assigning labels to data points is a component of classification jobs. The labels in the Live Yoga Pose Detection scenario would be the various yoga positions.

The use case diagram demonstrates how the various system blocks are utilized to carry out Live Yoga Pose Detection. Spyder is first opened by the system through Anaconda. After that, a GUI for the application is made with Spyder. Images of the user doing yoga poses are taken using the GUI. Spyder uses both image processing and video capturing methods to further process the images. The LR system then classifies the yoga positions based on the processed photos.

4. Results and Discussion

Figure 3. Softwares used

Spyder: Python's integrated development environment (IDE) is called Spyder. It is software that is accessible for Windows, macOS, and Linux and is free and open-source. Spyder is an effective tool for data analysis and scientific computing. It offers a simple interface for developing Python code and has a number of tools for debugging, profiling, and data visualization.

Python: Web development, data analysis, and machine learning are just a few of the many activities that can be accomplished with the general-purpose programming language Python. Because it is simple to learn and use and has a huge developer community, Python is a popular choice for creating programs that recognize yoga poses. This is because Python includes a number of libraries and tools for working with images and data.

SQLITE3: A compact, embedded relational database management system is SQLite3. It is a well-liked option for creating programs that detect yoga poses because it is simple to install and use and doesn't need a separate database server.

Anaconda: Anaconda is a Python and R distribution that is open-source and free. Spyder, Python, SQLite3, and a number of other tools for scientific computing and data analysis are included. The fact that Anaconda offers a complete environment for dealing with photos and data makes it a popular choice for those who want to create applications for detecting yoga poses.

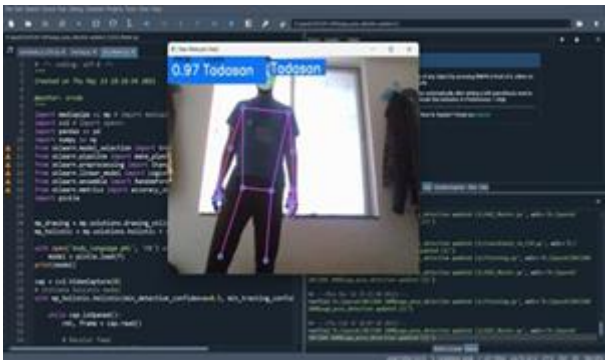

Figure 4. Working demonstration

The screenshot in Figure 4. shows a Python code that imports the following libraries:

cv2: The open-source, cost-free computer vision library is called OpenCV. It is utilized for computer vision, video analysis, and image processing jobs.

NumPy: The Python module NumPy is used for scientific computation. It is utilized for data analysis, mathematical calculations, and array manipulation.

pandas: Pandas is a Python data analysis library. It is utilized for data cleaning, manipulation, and visualization.

MediaPipe: A cross-platform, open-source framework for creating machine learning pipelines is called MediaPipe. Pose estimation, face detection, and other computer vision problems all make use of it.

5. Conclusion

Finally, in this project, we have achieved the following things:

- Successfully detected the yoga pose.
- Categorized the pose into four levels based on the average angle difference.
- Identified the erroneous part between the learner and the teacher.
- Measured the difference in body angles between an instructor's and a user's stance, as shown in Figure 4.

Successfully detected a number of Yoga Poses that are mentioned in Rig Veda from ancient history like Tanasan, Bhujangasan, Padmasana, and Trikonasana as shown in Figure 5.

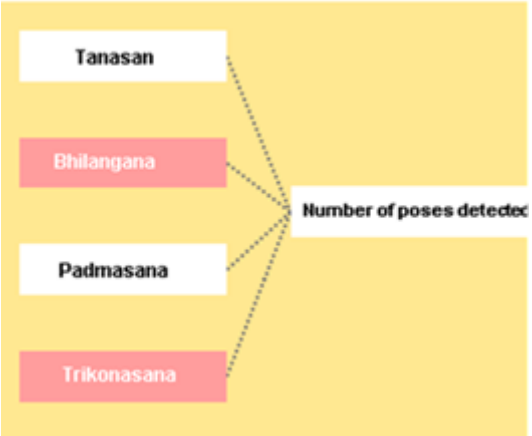

Figure 5. Number of Yoga Poses detected

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